

Indoor Air Quality Monitoring

Impact of FDM 3D Printing and Filtration in a Residential Setting

A 25-day self-hosted sensor study · Bambu Lab P2S · Sensirion SPS30 / SGP41

Executive Summary

This report analyses 25 days of continuous indoor air quality data collected with a custom-built monitoring station, alongside synchronised operating data from a Bambu Lab P2S 3D printer. The goal was to quantify how 3D printing — and the printer's VentoBox filtration — affect particulate matter (PM2.5) in a flat where the printer sits near the kitchen and a window.

Three findings stand out:

- Printing PLA or PETG had little measurable effect on PM2.5, with or without filtration. Within each period, particulate levels while printing were statistically indistinguishable from the room baseline when the printer was off.
- The apparent benefit of the VentoBox on PM2.5 could not be isolated. The room baseline itself was lower in the (later) VentoBox period than in the (earlier) no-filter period — even with the printer off — pointing to changing outdoor conditions rather than the filter. The filter's real value likely lies in gases and ultrafine particles, which this sensor does not measure (see Sections 4 and 7).
- The largest PM2.5 spikes were not caused by the printer. They cluster in the morning and evening and are consistent with cooking and/or outdoor air entering through the window — expected given the sensor's location.

Gas-phase measurements (VOC and NOx index) are excluded from this analysis because a firmware sampling issue prevented the Sensirion gas-index algorithm from calibrating. This has since been identified and fixed; gas data is a subject for future work (see Section 7).

1. Background & Objective

After upgrading from an old RF100 (PLA only) to a Bambu Lab P2S and beginning to print PETG, the motivation was simple: understand what is actually being breathed in a small flat during printing. Commercial air-quality monitors with charting and alerting were expensive, and inexpensive units have a reputation for unreliable readings — so a custom station was built around research-grade Sensirion sensors.

The system streams sensor data over MQTT to a self-hosted stack (Mosquitto, Telegraf, InfluxDB, Grafana) running on a Raspberry Pi. Crucially, the same pipeline also ingests live status from the Bambu Lab P2S over its local MQTT API, allowing air-quality readings to be correlated directly with printer activity, temperatures, and filtration settings.

2. Methodology

2.1 Hardware

- ESP32 (WROOM-32U, external antenna) — microcontroller and WiFi link
- Sensirion SPS30 — laser particulate matter sensor (PM1, PM2.5, PM4, PM10)

- Sensirion SGP41 — metal-oxide gas sensor (VOC / NO_x index)
- Bambu Lab P2S — print status via local MQTT (nozzle/bed temperature, print state, VentoBox fan speed)

2.2 Data collection

Readings were published every few seconds and aggregated to 5-minute means for analysis. The dataset spans 16 May to 11 June 2026 (25 days, 6,781 five-minute samples). Sensor and printer streams were time-aligned to the nearest sample. "Active printing" was defined as nozzle temperature above 150 °C, a reliable indicator of extrusion.

A small number of sensor-startup artefacts (simultaneous zero readings across all channels) were removed prior to analysis.

3. Overall Air Quality

Across the full monitoring period, indoor PM_{2.5} was generally good by WHO standards:

Metric	Value
Mean PM _{2.5}	5.7 µg/m ³
Median PM _{2.5}	4.5 µg/m ³
95th percentile	12.8 µg/m ³
Maximum (5-min mean)	372 µg/m ³
Time above WHO 24h guideline (15 µg/m ³)	3.0%
Time above alarm threshold (35 µg/m ³)	0.31%

For context, the WHO 2021 air quality guideline for PM_{2.5} is an annual mean of 5 µg/m³ and a 24-hour mean of 15 µg/m³. The flat sat below the 24-hour guideline 97% of the time. The rare excursions are examined in Section 5.

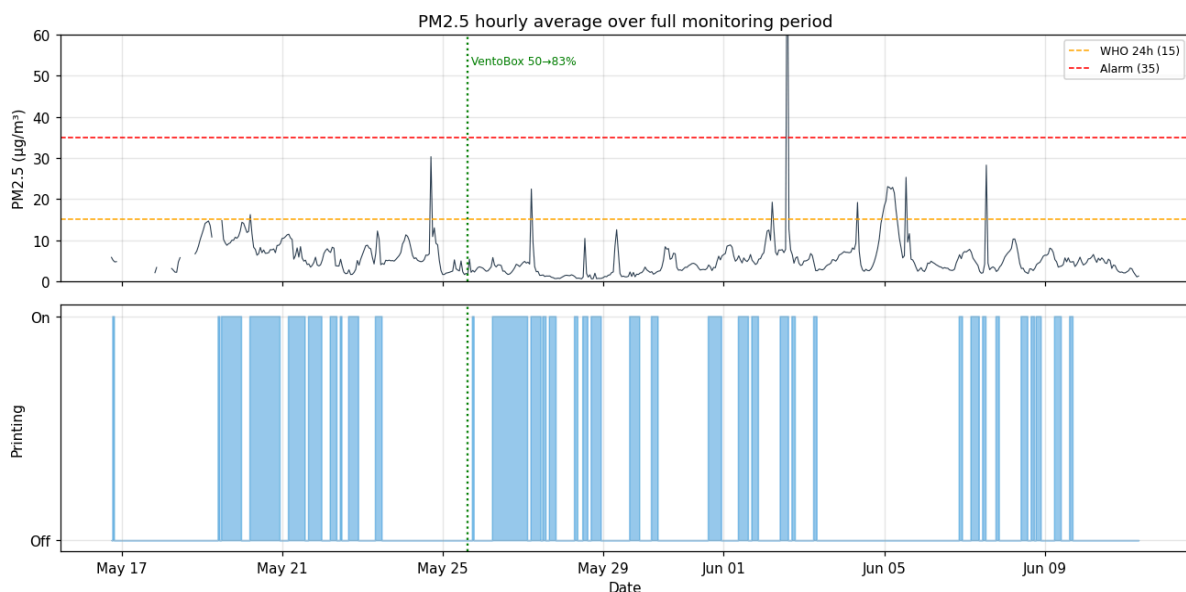


Figure 1. Hourly-averaged PM_{2.5} over the full period (top) with printing activity (bottom). The green dotted line marks the VentoBox installation on 25 May. Orange and red dashed lines are the WHO 24-hour guideline and the chosen alarm threshold.

4. The VentoBox: What the Data Can and Cannot Show

The VentoBox filtration unit was installed on 25 May. Before that date the printer ran with no filtration. A natural question is whether the filter reduced printing-related particulates. At first glance the data appears to say yes: median PM_{2.5} during printing was 6.7 $\mu\text{g}/\text{m}^3$ before installation and 2.9 $\mu\text{g}/\text{m}^3$ after. However, this comparison is confounded and does not support a causal claim.

The problem is that the room's background air quality also changed between the two periods — for reasons unrelated to the printer or the filter. Looking only at times when the printer was switched off, the median room baseline was 6.1 $\mu\text{g}/\text{m}^3$ during the no-filter period (mid-May) and 3.6 $\mu\text{g}/\text{m}^3$ during the VentoBox period (late May to June). Since the printer (and therefore the VentoBox) was off during these measurements, the filter cannot explain the drop. The most likely cause is changing outdoor conditions — mid-May included several visibly polluted days (daily baselines of 12–13 $\mu\text{g}/\text{m}^3$ on 19–21 May) while early June was cleaner.

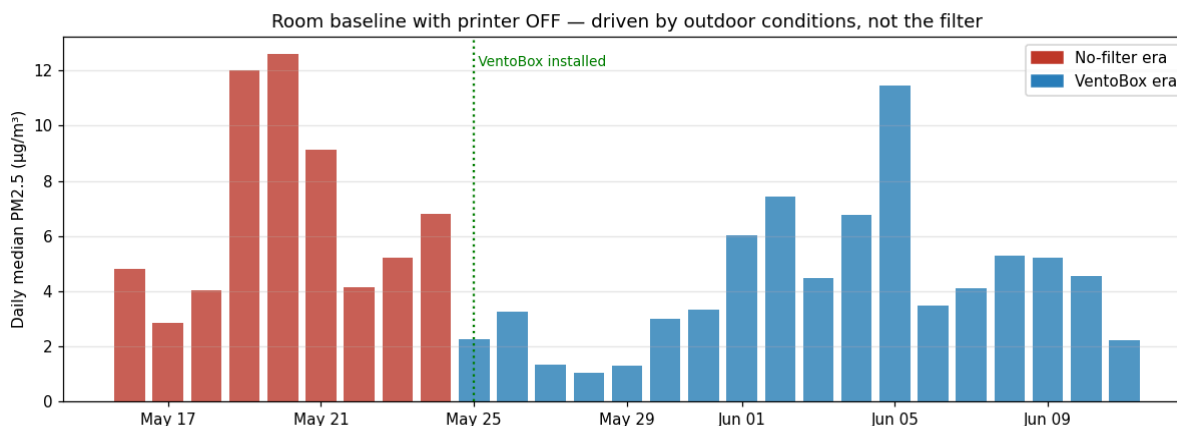


Figure 2. Daily median PM_{2.5} with the printer switched off. Because the printer and filter were inactive, this curve reflects the room's background air — which was higher in mid-May and lower in June, independent of the VentoBox.

The honest way to assess the printer's impact is to compare printing against the room baseline within the same period, so that outdoor conditions are held roughly constant. Doing this shows printing had a minimal effect in both periods:

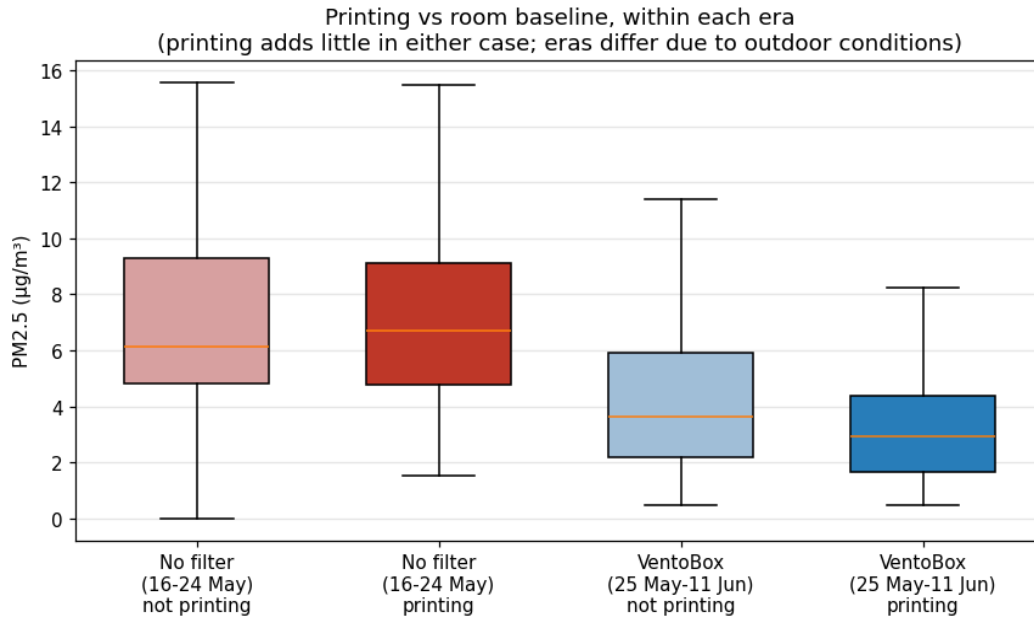


Figure 3. PM2.5 while printing versus the room baseline, within each period. In both cases the printing and not-printing distributions overlap heavily; the visible shift is between periods (outdoor conditions), not between printing and idle.

In the no-filter period, printing raised median PM2.5 by just +0.6 $\mu\text{g}/\text{m}^3$ over baseline — not statistically significant (Mann-Whitney $p = 0.18$). In the VentoBox period, printing was actually slightly below baseline. In neither case did printing meaningfully degrade the air.

Conclusion on the VentoBox: this dataset cannot demonstrate a PM2.5 benefit from the filter, because printing barely raised particulates even without it. This does not mean the VentoBox is useless. Two important caveats apply. First, the SPS30 measures particles down to roughly 300 nm and does not capture the ultrafine particles (UFPs, <100 nm) that dominate FDM emissions — a HEPA stage can still help here. Second, the filter's activated-carbon layer targets gases (VOCs, styrene), which were not measured in this study. The filter may well provide benefits this particulate sensor is simply blind to.

5. PLA vs PETG: Does Material Matter?

Although filament type was not logged for each print, nozzle and bed temperatures act as a reliable proxy. The printing data fell into two clean clusters: a low-temperature group (nozzle ~220 °C, bed 55 °C) consistent with PLA, and a high-temperature group (nozzle 245–255 °C, bed 70 °C) consistent with PETG. Using these signatures, 93% of printing samples could be classified confidently.

A fair comparison requires controlling for confounders: PLA was printed more often before the VentoBox was installed, and the dataset contains occasional external pollution spikes unrelated to printing. The comparison below is therefore restricted to the period after VentoBox installation, with external events ($\text{PM}_{2.5} > 50 \mu\text{g}/\text{m}^3$) excluded.

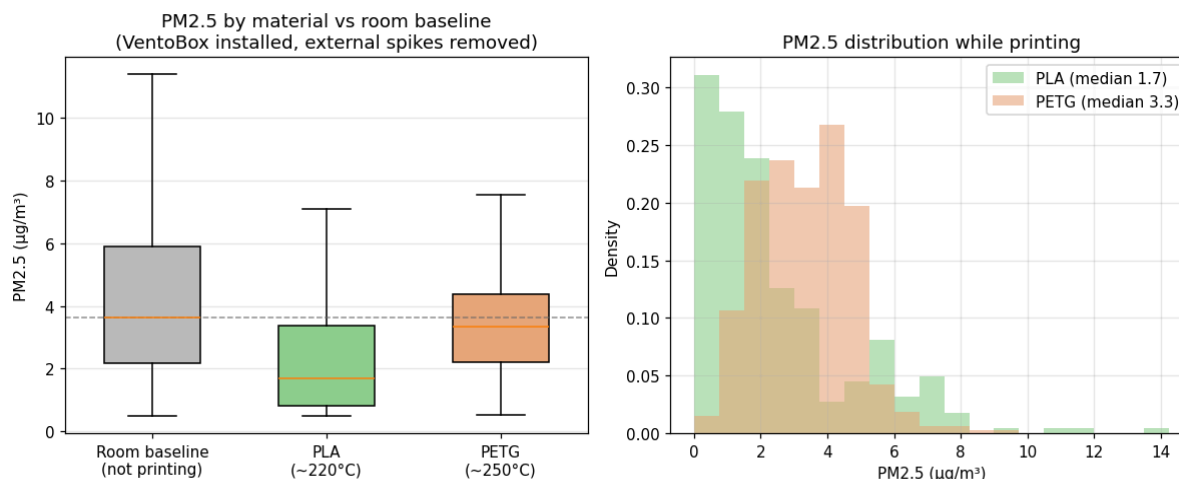


Figure 4. PM2.5 by material versus the room baseline (VentoBox installed, external spikes removed). PETG prints show roughly double the median PM2.5 of PLA, but both stay at or below the non-printing room baseline (dashed line).

The result has two parts. First, PETG does emit more particulates than PLA: its median PM2.5 during printing (3.3 $\mu\text{g}/\text{m}^3$) is about twice that of PLA (1.7 $\mu\text{g}/\text{m}^3$), a highly significant difference ($p \approx 2 \times 10^{-18}$). This matches expectations, since PETG is extruded around 30 °C hotter. Second — and more reassuring — with the VentoBox running, neither material pushed PM2.5 above the normal room baseline (3.6 $\mu\text{g}/\text{m}^3$). PLA printing actually sat about 1.9 $\mu\text{g}/\text{m}^3$ below baseline and PETG about 0.3 $\mu\text{g}/\text{m}^3$ below it.

In other words, the difference between PLA and PETG is real but small in absolute terms, and the filtration is effective enough that printing either material did not measurably degrade the air relative to the room's usual state. The higher PETG figure is best read as “less strongly filtered below baseline,” not as a hazardous level.

6. What Actually Drives the Spikes

Counter to initial expectations, the printer was not the main source of high PM2.5 readings. Of the ten highest 5-minute readings, eight occurred while the nozzle was cold (not printing). The top five events:

Time (UTC)	PM2.5	Printing?
2 Jun 14:30	372 $\mu\text{g}/\text{m}^3$	Yes
2 Jun 14:35	300 $\mu\text{g}/\text{m}^3$	No
2 Jun 14:40	240 $\mu\text{g}/\text{m}^3$	No
24 May 17:55	185 $\mu\text{g}/\text{m}^3$	No
5 Jun 13:10	171 $\mu\text{g}/\text{m}^3$	No

The single highest reading does coincide with printing, but it is immediately followed by even larger readings after the nozzle cooled — suggesting an external event (e.g. cooking, a window opened to a smoggy street) that happened to overlap the end of a print, rather than being caused by it.

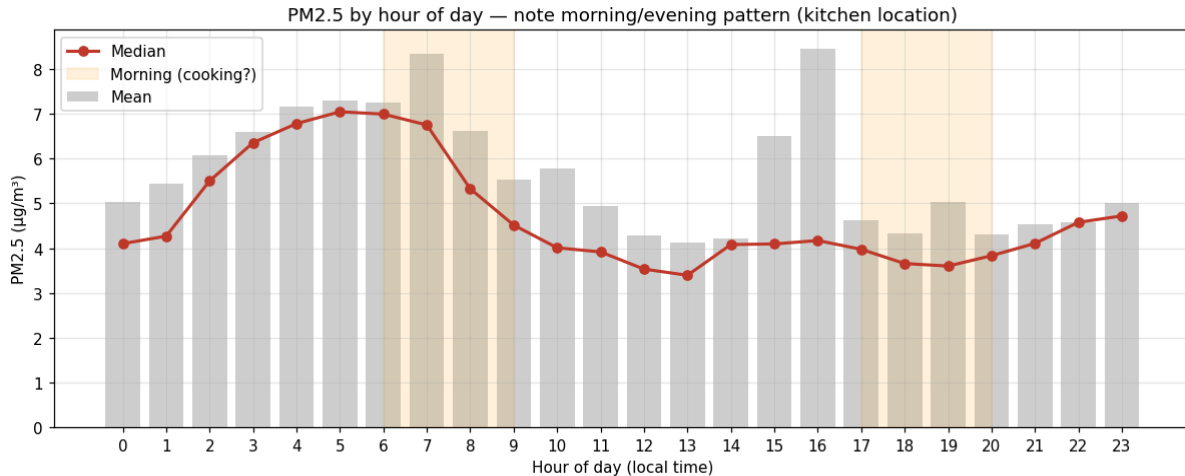


Figure 5. PM2.5 by hour of day (local time). The pronounced early-morning and evening pattern is characteristic of cooking and/or outdoor air ingress — consistent with the sensor's placement near the kitchen and a window — rather than printing, which occurred at varied hours.

Because the sensor sits near the kitchen and a window, two plausible external sources are cooking emissions and outdoor air (including seasonal smog). Distinguishing the two would require an outdoor reference sensor and is left for future work.

7. Limitations & Future Work

7.1 Gas-phase data (VOC / NOx)

VOC and NOx index values were recorded but are not analysed here. Throughout the dataset they remained pinned at their initialisation values (VOC = 1, NOx = 100) with zero variance, even across a continuous 30-hour window. Investigation showed this was not a hardware fault: the Sensirion gas-index algorithm requires its processing function to be called at a steady 1 Hz, but the original firmware called it irregularly (skipped cycles when the PM sensor was not ready, plus variable timing from network operations), so the algorithm never built its internal baseline.

The firmware has been rewritten to sample the gas sensor on a dedicated, unconditional 1-second timer, independent of the particulate sensor and network activity. Once fresh data accumulates, VOC/NOx — including filament off-gassing and the isopropyl-alcohol response observed live on the serial monitor — can be analysed properly.

7.2 Device reliability

Diagnostic logging (free heap, RSSI, CPU temperature, boot reason) was used to characterise stability. Over the period the device restarted on 29 occasions. Boot-reason analysis showed an early cluster (18–19 May) of power-cycles and WiFi-loss restarts during active development, followed by periodic interrupt-watchdog resets every 1–3 days. Free heap was rock-stable throughout (never below 219 KB, drift ≈ -5 bytes/hour), ruling out a memory leak; the watchdog resets are attributed to occasional blocking on the I2C bus or network calls. The revised firmware adds an I2C timeout, bounded reconnect attempts, and regular watchdog servicing to address this.

7.3 Other limitations

- No outdoor reference sensor, so indoor-vs-outdoor attribution of spikes is inferred, not measured.
- Filament type (PLA vs PETG) was not logged per-print, so material-specific emission differences could not be isolated.
- 5-minute averaging smooths short transient events; high-resolution capture would better resolve brief emission peaks.

8. Conclusions

- Indoor PM2.5 in the flat was good overall, below the WHO 24-hour guideline 97% of the time.
- Printing PLA or PETG barely raised PM2.5 above the room baseline — even without filtration (+0.6 $\mu\text{g}/\text{m}^3$, not significant). A clear PM2.5 benefit from the VentoBox could not be isolated, because the printer added so little to begin with.
- The VentoBox should not be dismissed: this study only measured particles above ~300 nm and no gases, so the filter's benefits for ultrafine particles and VOCs were outside what the sensor could detect.
- PETG produced about twice the median PM2.5 of PLA while printing (3.3 vs 1.7 $\mu\text{g}/\text{m}^3$), consistent with its higher extrusion temperature — but with the VentoBox running, both materials stayed at or below the room baseline.
- The largest pollution events originated outside the printer, consistent with cooking and outdoor air given the sensor location.
- Gas-phase monitoring is pending a firmware fix that has now been implemented; this is the primary direction for follow-up analysis.

Data collected 16 May – 11 June 2026. Analysis based on 6,781 five-minute samples of sensor data time-aligned with 4,652 printer status records. Hardware: ESP32 + Sensirion SPS30/SGP41; pipeline: MQTT → Telegraf → InfluxDB → Grafana.